Auctions
Beyond single-item

COMP6203 Intelligent Agents

Dr Baharak Rastegari
b.rastegari@soton.ac.uk
Electronics and Computer Science
University of Southampton

Learning Outcomes 1

By the end of this lecture, you should be able to

- **Define** multi-unit auctions and combinatorial auctions settings
- **Describe** three different auctions for multi-unit single-unit demand setting
- **Compute** the outcome (allocation and payment) or running any of these three multi-unit single-unit demand auctions
- **Describe** the Revelation Equivalence theorem for multi-unit case and explain its implications

Previously

- Learned about single-item auctions
  - One item is put up for sale
  - Information from this sale is not carried forward to future auctions
- Made some simplifying assumptions
  - Independent private value model
  - Bidders cannot collude
- Set the objective be maximising the social welfare
  - Award to the bidder with the highest value

Recall: Sealed-bid single-item auctions

- First-price auction
  - Shade your bid: put a bid lower than your valuation.
  - How much lower? Do the math!

- Second-price (Vickrey) auction
  - Placing a bid equal to your valuation is a dominant strategy → Vickrey auction is (dominant strategy) truthful.

Multi-unit auctions with single-unit demand

- There are *k* identical items for sale in a single auction (i.e., one auction)
- There are *n* strategic risk-neutral bidders
- Each bidder only wants one unit
- Bidders have independent private values for these single units

- What auction should we run?
  - Which bidders should each be allocated one of the items
  - How much should each of the winners pay

Your thoughts

Let’s start with 2 items

- There are 2 identical items (say first edition of a book) for sale in a single auction
- There are *n* strategic bidders
- Every bidder only wants one unit

- What auction should we run?
  - Which two bidders should each be allocated one of the items
  - How much should each of the winners pay

VCG: Vickrey extended to beyond single-item

Assume there are *k* identical items for sale, each bidder wants one unit

- The top *k* bidders each win one item
- The payment of each winning bidder is equal to the k+1st highest bid

- Is VCG efficient? I.e., does it maximise social welfare?
- Is VCG (dominant strategy) truthful?

Example

What is the revenue of VCG when:

- $k = 1$
  - £20
- $k = 2$
  - £30
- $k = 3$
  - £24
- $k = 4$
  - £0

Bidder	Bid amount
1	£25
2	£20
3	£15
4	£8

Some other auctions

- Pay-your-bid: Discriminatory pricing, because bidders will pay different amounts for the same thing.
  - ❖ Generalises first-price auction
- Lowest winning bid: Similar to VCG but ensures that bidders don't pay zero if there are fewer bids than units for sale.
  - Used in eBay
  - Is it truthful?
  - How else can we ensure bidders don't pay zero if there are fewer bids than units for sale, but ensure truthfulness?
- Sequential auctions: repeated single-item auctions Next Lecture

Recall: Revenue Equivalence

Assume that all bidders are risk-neutral and utility maximisers, and each has an independent private valuation for the single item, drawn from a common cumulative distribution $F(\theta)$ that is strictly increasing and atomless on $[L,H]$. Then any auction mechanism in which

- in equilibrium, the item will be allocated to the agent with the highest valuation, and
- any agent with valuation L has an expected utility of zero;

yields the same expected revenue, and hence results in any bidder with valuation θ making the same expected payment.

First-price, second-price, English and Dutch auctions all satisfy the above. Hence, they are **revenue equivalent**.

Revenue Equivalence for multi-unit auctions

Assume that all bidders are risk-neutral and utility maximisers, and each has an independent private valuation for the single item, drawn from a common cumulative distribution $F(\theta)$ that is strictly increasing and atomless on $[L,H]$. Then any auction mechanism in which

- in equilibrium, the items are allocated efficiently (i.e. the k bidders with the highest valuation each win one item), and
- any agent with valuation L has an expected utility of zero;

yields the same expected revenue, and hence results in any bidder with valuation θ making the same expected payment.

All payment rules discussed earlier (VCG, pay-your-bid, lowest winning bid) satisfy the above. Hence, they are **revenue equivalent**.

- If assumptions of the theorem hold: bidders are risk-neutral and play equilibrium strategies

(general) Multi-unit auctions

- There are *k* identical items for sale (e.g. general admission ticket to a concert) in a single auction (i.e., one auction)
- There are *n* strategic risk-neutral bidders
- Each bidder wants one or more units
- Bidders have independent private values for these single units

- What auction should we run?
  - Which bidders should each be allocated one of the items
  - How much should each of the winners pay

Example: concert tickets

- Five tickets for sale, 4 bidders interested in purchasing 1 or 2 tickets
- What allocation is efficient (social welfare maximizing)?
  - Bidders 1 and 2 each get 2 tickets, bidder 3 gets 1 ticket

Bidder	Valuation for 1 ticket	Valuation for 2 tickets
1	£25	£50
2	£20	£40
3	£15	£30
4	£8	£16

- What should each winner pay for the mechanism to be dominant strategy truthful?
  - Bidder 1 pays £(20*2+15*2+8)-£(20*2+15) = 78-55= £23
  - Bidder 2 pays £(25*2+15*2+8)-£(25*2+15) = 88-65= £23
  - Bidder 3 pays £(25*2+20*2+8)-£(25*2+20*2) = 98-90= £8

Bidder 1 and 2 pay £11.5 per ticket, Bidder 3 pays £8 for a ticket

VCG for general multi-unit auctions

Assume there are *k* identical items for sale, each bidder wants one or more units

- No longer k+1st-price auction
- All winning bidders who won the same number of units with pay the same amount as each other
- Bidders who win different numbers of units will not necessarily pay the same per unit prices
- Bidders who win large number of units will pay at least as much in total (not necessarily per unit) as bidders who won smaller numbers of units

Learning Outcomes 2

By the end of this lecture, you should be able to

- **Define** sequential auctions and combinatorial auctions settings
- **Compute** the optimal strategy in a sequential auction when you know the probability distribution of opponent bids (as in a decision theoretic framework)

Revisit Example: concert tickets

- Five tickets for sale, 4 bidders interested in purchasing 1 or 2 tickets
- Efficient allocation gives bidder 1 and 2 each two tickets, and bidder 3 one ticket
- Why charging each winner the highest losing bid (here £15) per awarded ticket is not truthful?

Bidder	Valuation for 1 ticket	Valuation for 2 tickets
1	£25	£50
2	£20	£40
3	£15	£30
4	£8	£16

Multi-unit Valuations (K units of the same item)

How can bidders express their valuation in a multi-unit auction?

Let S denote some set of items

- General: let $p_1, p_2, \ldots p_k$ be arbitrary, nonnegative real numbers. Then $\theta(S) = \sum_{j=1}^{|S|} p_j$ (p_j denotes how much bidder values obtaining the jth unit)
- Downward sloping: general, but $p_1 \geq p_2 \geq \cdots \geq p_k$
- Single-item: same value for any bundle of size at least 1
- Additive: $\theta(S) = c|S|$ for some constant c (ticket example)
- Fixed budget: similar to the additive, but the bidder has a budget B, so that $\theta(S) = \min(c|S|, B)$
- Majority: $\theta(S) = c$ if $|S| \geq \frac{k}{2}$, 0 otherwise

Sequential Auctions

- There are *k* identical items for sale (e.g. a dozen new cars), in a sequence of *k* single-item auctions
- There are *n* strategic risk-neutral bidders
- Each bidder only wants one unit
- Bidders have independent private values for these single units

- A bidding strategy for each bidder maps to a tuple of size *k*, specifying the bidder's bid for each of the *k* auctions:
  - $(s_i^1, s_i^2, \ldots, s_i^k)$ where $s_i^t : \Theta \to A$

Decision theoretic framework : recall

- The bidder have beliefs about bid distribution
  - In case of ties, the bidder loses
  - F(b) denotes the probability that all other bids are less than b. So, F(b) is the probability of winning given bid b.

- Single auction: Goal is to find bidding strategy $s: \Theta \rightarrow A$, where A is the set of actions or bids, which maximizes the expected utility for any valuation $\theta_i \in \Theta$.
- Sequence of k single auctions: Goal is to find bidding strategy $\left(s_i^1, s_i^2, \ldots, s_i^k\right)$, $s_i^t: \Theta \rightarrow A$, which maximizes the expected utility for any valuation $\theta_i \in \Theta$.

Recall: First-price example from Auctions lecture

- Suppose bids are ranging from 0 to 5, bidder i's true value is 4 and they have the following information (same as Auctions lecture)

  F(0) = 0, F(1) = 0.2, F(2) = 0.7, F(3) = 0.8, F(4) = 0.9, F(5) = 1.0

- We calculated the expected uiltiy of bidder i for the different bids they can place:

$E(u_i(4)|4] = 0, E(u_i(4)|3] = 0.8, E(u_i(4)|2] = 1.4, E(u_i(4)|1] = 0.6, E(u_i(4)|0] = 0$

- We concluded that i's optimal bid is **2**

Example: sequence of 2 First-price auctions

- You have 2 units of the same item, selling them sequentially using First-price auction
- Given the same information about F(b) as in the previous slide, what is the optimal strategy for bidder i ? i.e. what pair of bids $\left(b_i^1, b_i^2\right)$ maximise bidder i's utility?

F(0) = 0, F(1) = 0.2, F(2) = 0.7, F(3) = 0.8, F(4) = 0.9, F(5) = 1.0

$E(u_i(4)|4] = 0$, $E(u_i(4)|3] = 0.8$, $E(u_i(4)|2] = 1.4$, $E(u_i(4)|1] = 0.6, E(u_i(4)|0] = 0$

Another example:

- Suppose bids are ranging from 0 to 8, bidder i's true value is 6 and F(b) = b*0.125
- Expected utility of bidder i for different bids in a single First-price auction:
- Optimal bid is 3

b	F(b)	$\theta_i - b$	$E[u_i(6)\|b]$
0	0	6	0
1	0.125	5	0.625
2	0.25	4	1
3	0.375	3	1.125
4	0.5	2	1
5	0.625	1	0.625
6	0.75	0	0.75
7	0.875	-1	-0.875
8	1	-2	-2

Example contd.

- Now assume there are 2 identical items sold in a sequence of 2 First-price auctions
- If the bidder loses in the first auction, they will bid in the second auction (if they win in the first auction, they don't show up to the second auction)
- What pair of bids $\left(b_i^1, b_i^2\right)$ maximise bidder i's utility?
  - If they reach the second auction, optimal strategy is for i to bid 3, achieving utility of 0.125
  - What should i bid in the first auction?

Example contd.

b	F7(b)	$\theta_i - b$	$E[u_i(6)\|b]$	1-F(b)	$E[u_i(6)$ $from\ second$ $auction\|b_i^2 = 3]$
0	0	6	0	1	
1	0.125	5	0.625	0.875	
2	0.25	4	1	0.75	
3	0.375	3	1.125	0.625	
4	0.5	2	1	0.5	
5	0.625	1	0.625	0.375	
6	0.75	0	0.75	0.25	
7	0.875	-1	-0.875	0.125	
8	1	-2	-2	0	

Discussion on sequential auctions

- Bidders' bids increase from one auction to the next
  - Intuition: bidders have positive expected utility after losing earlier auctions, because they can participate in future rounds. As the number of rounds decrease, so does this expected utility.
- In the decision theory framework, last point implies that F is expected to increase from one auction to the next
- Putting game theoretic hat on, it can be shown that in equilibrium
  - Bidders' bid increase from one auction to the next
  - In a give auction, bidders with higher valuation place higher bids

Combinatorial auctions

- Multiple, heterogenous items are being sold
- Consider the sorts of valuations that agents could have in this case
  - Complementarity: for disjoint subsets S and T of items
    $$\theta(S \cup T) > \theta(S) + \theta(T)$$
    - E.g., a left shoe and a right show
  - Substitutability: for disjoint subsets S and T of items
    $$\theta(S \cup T) < \theta(S) + \theta(T)$$
    - E.g. two tickets to different concerts at the same time
- Substitutability is relatively easy to deal with
  - E.g. sell goods sequentially, or allow bid withdrawal
- Complementarity is trickier

Think about the maritime shipping

Example on combinatorial auction

Discussion on combinatorial auctions

Fun Game

1	2	3
4	5	6
7	8	9

- 9 plots of land for sale, geographically related as shown
- Independent private values v_i, uniformly distributed between 2 and 10
- Valuation for a set S
  - if you get one plot other than #5: v_i
  - Plot #5: $2v_i$
  - Any two plots: $3v_i$
  - Any three or more plots: $5v_i$
- Rules:
  - Auctioneer moves from one plot to the next sequentially, holding an English auction for each plot
  - When bidding stops on a plot, move on to the next plot
  - After bidding stops for plot #9, auction ends